Our mission

Green Capital Allocation & Portfolio Intelligence

Direct capital toward evidence, not assertions.

Combine market intelligence and climate evidence to balance risk and return under explicit investment rules.

How can AI make sustainable investment decisions more verifiable?

Evidence
Source-aware ESG assessments

Intelligence
Dated market signals and risk

Accountability
Constrained allocation + audit trail

Implemented as a decision-support simulator.

The problem

Financial upside and sustainability credibility can point in different directions.

Evidence is fragmented

Issuer claims, price histories and policy rules sit in different places.

Missing evidence must stay unverified.

Rare events hide failure

A 99% accurate model can miss every positive case in a 99:1 split.

Accuracy alone is the wrong proof.

Mistakes have unequal costs

False alarms consume audit time. Missed events can create financial loss.

The operating threshold needs economics.

The need: a decision that a reviewer can inspect, challenge and reproduce.

Our solution: three roles, one auditable decision

LangGraph coordinates the work; deterministic constraints govern the allocation.

ESG Verifier

Issuer-scoped evidence
Citations and validated output
No evidence → unverified

Market Analyst

Cached adjusted prices
Historical return and volatility
KRBN ETF proxy context

Portfolio Optimizer

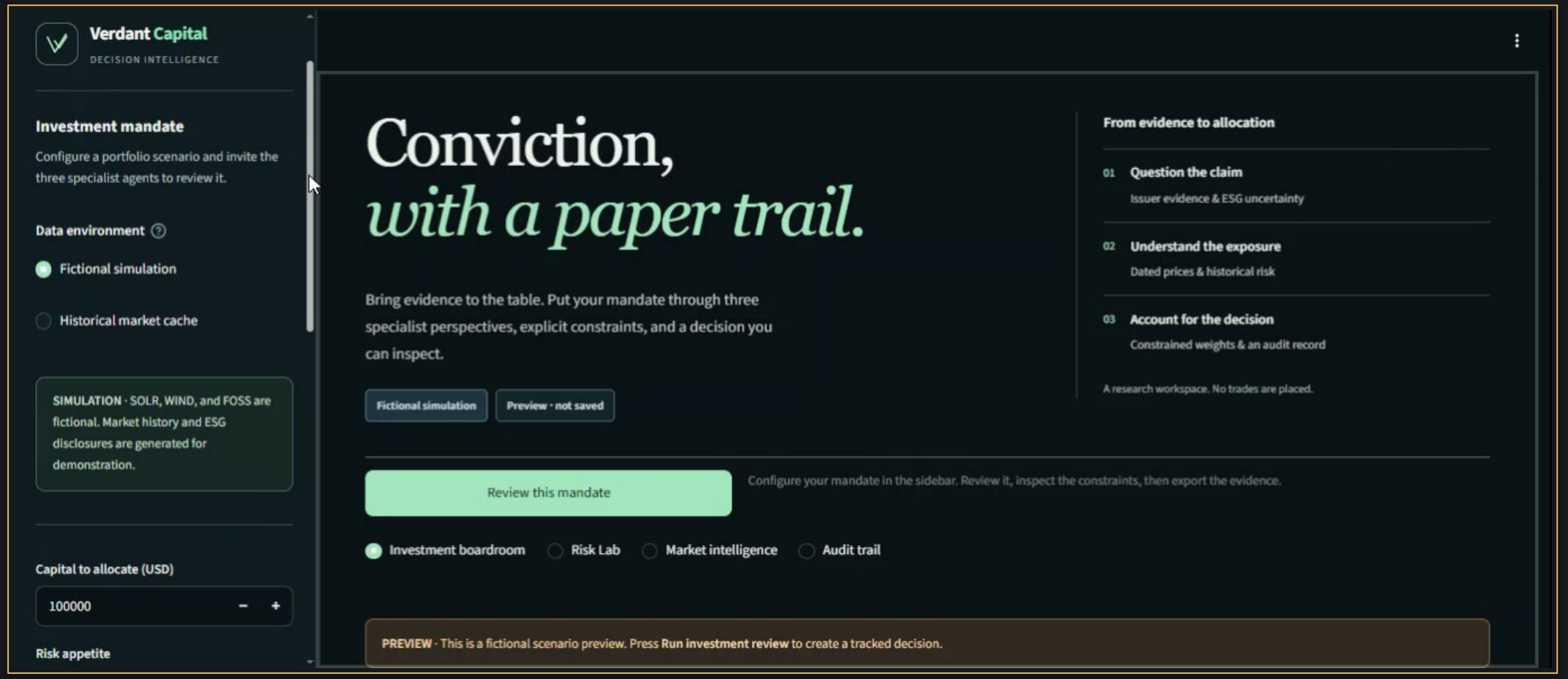
SciPy SLSQP weights
Long-only + ESG exposure caps
Unused capital stays in cash

$$\text{Minimize } \lambda/2 \cdot w^{\top}\Sigma w - \tilde{\mu}^{\top}w \qquad w \geq 0 \quad \Sigma w = 1 \quad w_i \leq \text{cap}_i$$

Bounded conflict review → deterministic memo → SQLite record + JSON export

Product walkthrough

Click the video to play in desktop PowerPoint · 62-second silent screen recording



Supplied recording uses the earlier mint theme; the deck and current interface use charcoal/gold.

C02 + C06: detect rare cases, explain the mistakes

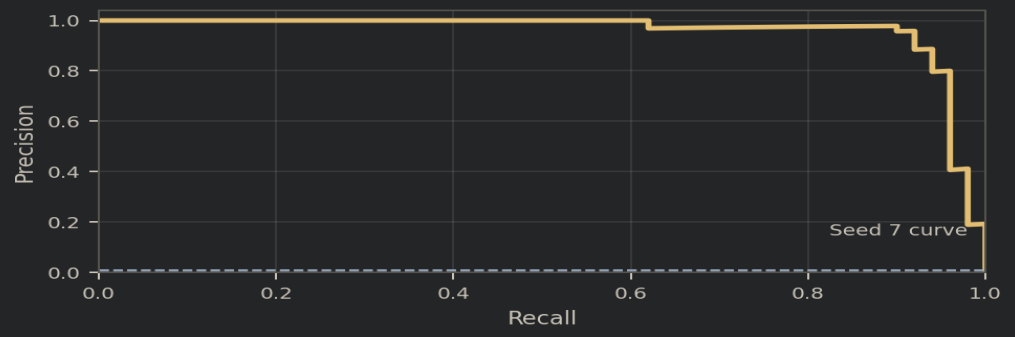
Controlled synthetic evidence; not validated real-company fraud or default detection.

C02 / 99:1 rare-event stress

0.9583

Mean PR-AUC across five seeds

4,950 negatives + 50 positives per test split



C06 / three held-out SHAP cases

TRUE POSITIVE +0.0014

No third-party verification

FALSE POSITIVE +0.0017

Missing baseline

FALSE NEGATIVE −0.0120

Missing baseline

Base score + Σ SHAP = model score

Bars show the dominant feature; full plots are in the report.

C13: optimize the cost of a decision

Select on validation data. Freeze the threshold. Evaluate on untouched test cases.

Cost matrix / USD

	No audit	Audit
Actual negative	TN **$0**	FP **$100**
Actual positive	FN **$10,000**	TP **$100**

(TP + FP) × $100
+ FN × $10,000

Held-out confusion

	Not flagged	Flagged
	TN **4,920**	FP **30**
	FN **1**	TP **49**

Threshold −0.0315983

79 audits · 98% recall

DEFAULT TEST LOSS

$35,200

VALIDATION-SELECTED POLICY

$17,900

$17,300 less loss

Illustrative economics, not realized savings.

Thank you.

Evidence-led capital. Auditable decisions.

Capital should follow evidence.

A working research prototype today.
Independent issuer evidence and real-world validation next.

Questions?

github.com/syedazoh/verdant_capital-